SESSION 1

Opening & Introduction to GeoAI

Welcome · What is GeoAI · How It Works · Data Types · Applications & Case Studies · The GeoAI Canvas · Why Innovation Matters

Day 1 · 7 July 2026 · GeoAI Hackathon Programme

How to use these notes. *These notes are written to be self-contained: everything covered in the live session — plus extra background, diagrams, worked examples, a glossary, and self-check questions — is included, so you can follow the full content even if you could not attend.*

Learning Objectives

By the end of this session, you should be able to:

- Define GeoAI, explain how it differs from traditional GIS, and describe how a GeoAI system works end-to-end.
- Recognise the six main types of geospatial data and what each enables.
- Name the most common GeoAI tasks (feature extraction, classification, detection, prediction) and give an example of each.
- Describe real GeoAI applications across climate, smart cities, agriculture, and disaster management, with Malaysian examples.
- Explain why innovation and cross-disciplinary collaboration are essential for GeoAI projects.
- Complete a Problem–Data–Solution Canvas for a real problem in your own community (Activity 1).

1. What Is GeoAI?

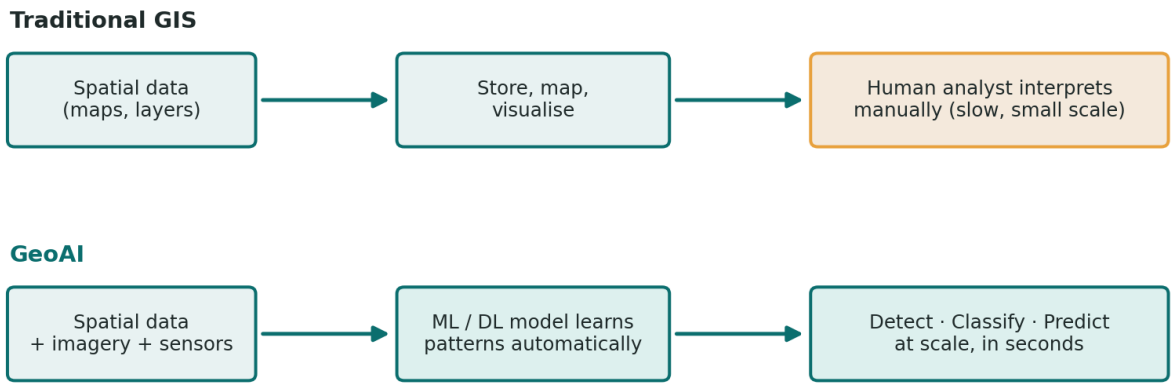
Definition. GeoAI (Geospatial Artificial Intelligence) is the fusion of AI methods — machine learning, deep learning, and computer vision — with location-based data such as satellite imagery, drone footage, sensors, and GPS traces, in order to detect change, predict risk, and guide real-world decisions.

Put simply, GeoAI sits at the intersection of artificial intelligence and location intelligence. Industry leader Esri describes it as the integration of geospatial data with AI techniques — machine learning, deep learning, and increasingly large language models — to extract features, uncover patterns, make predictions, and automate spatial workflows at a scale and speed that manual analysis cannot match. The term “GeoAI” itself was coined at the first ACM SIGSPATIAL GeoAI Workshop in 2017 and the field has grown rapidly since.

The Two Engines Inside GeoAI: Machine Learning and Deep Learning

- **Machine Learning (ML)** means training a computer with data-driven algorithms so it can recognise patterns in geospatial data and predict outcomes — without being given explicit rules. Classic ML methods used in GeoAI include Random Forests (used in the KL heat-island study below), clustering, regression, and time-series forecasting.
- **Deep Learning (DL)** is a subset of ML that uses many connected neural-network layers — loosely inspired by the human brain — to automatically extract progressively more complex features from data. DL is especially powerful for imagery: it can learn to spot a building, a flooded road, or a stressed oil palm directly from pixels, and for spatial processes whose mechanics are not fully understood.

Traditional GIS vs GeoAI



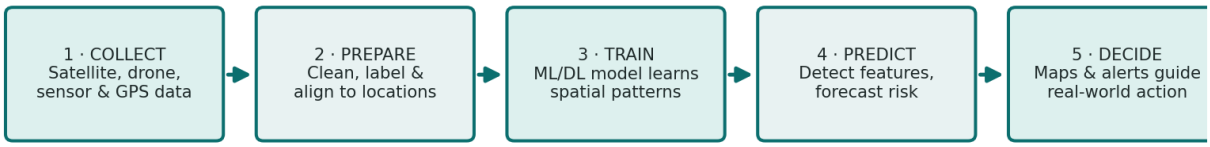
The shift: GIS shows the map — GeoAI reads the map for you.

Figure 1.1 — Traditional GIS relies on manual human interpretation; GeoAI automates pattern detection and prediction at scale.

System	What it does
Traditional GIS	Stores, maps and visualises spatial data for people to interpret manually.
GeoAI	Automatically detects patterns, classifies features, and predicts outcomes at scale.

Important: GeoAI does not replace GIS — it supercharges it. In practice, GeoAI output (e.g. extracted building footprints) usually becomes a new input layer for further GIS analysis.

How a GeoAI System Works, End to End



A feedback loop: real-world outcomes become new training data for the next model version.

Figure 1.2 — The five-step GeoAI workflow, from raw data to real-world decisions.

1. **Collect:** gather location-based data — satellite and drone imagery, LiDAR and sensor networks, GPS and mobility traces, crowdsourced map data, historical GIS records.
2. **Prepare:** clean, label, and align the data to locations. In practice this is often the most time-consuming step — models are only as good as their training data.
3. **Train:** an ML or DL model learns the spatial patterns that link inputs (e.g. rainfall, elevation) to outcomes (e.g. flooding). Teams rarely start from zero: platforms such as ArcGIS offer 50+ pretrained deep-learning models for tasks like building-footprint extraction, road digitisation, and land-cover classification, which can be used out of the box or fine-tuned with local data.
4. **Predict:** the trained model detects features, classifies pixels, or forecasts risk on new data — in seconds, across whole regions.
5. **Decide:** predictions become maps, dashboards, and alerts that guide real decisions — where to send help, what to plant, when to evacuate. Outcomes then feed back as new training data.

Trend to know (2025–26): the field is moving toward geospatial foundation models — very large neural networks (e.g. Prithvi, Clay) pretrained on vast collections of satellite, radar, and LiDAR imagery, which teams then fine-tune for specific tasks. For a hackathon, though, small well-tuned models and pretrained task models are usually the right choice.

The Most Common GeoAI Tasks

Task	What it means · example
Feature extraction	Pull discrete objects out of imagery — e.g. extracting every building footprint or road segment from aerial photos.
Pixel classification	Assign every pixel a category — e.g. land-cover maps (forest / water / built-up / cropland).
Object detection	Find and box specific objects — e.g. detecting wind turbines, vehicles, or palm trees in satellite images.
Change detection	Compare imagery over time to flag what changed — e.g. new deforestation next to a plantation.
Prediction / regression	Forecast a value or risk — e.g. flood risk in the next 6 hours, or crop yield this season.
Clustering & pattern mining	Discover natural groupings — e.g. neighbourhoods with similar heat-exposure profiles.

2. The Six Types of GeoAI Data

Know these six data types going into the hackathon — they determine what a GeoAI model can and cannot do with your idea. Most strong projects combine at least two or three of them.

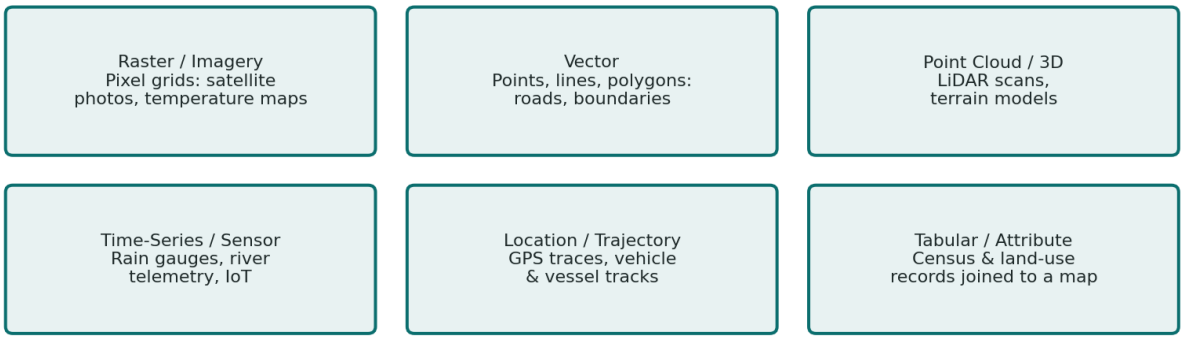


Figure 1.3 — The six families of geospatial data.

Data type	What it is	Examples
Raster / Imagery	Pixel grids representing continuous surfaces — brightness, elevation, or temperature per cell.	Satellite photos; aerial imagery; land-surface temperature maps
Vector	Discrete features drawn as points, lines and polygons, each with attributes attached.	Building footprints; roads; plantation or admin boundaries
Point Cloud / 3D	Dense XYZ measurements capturing surface shape, height, and structure in 3D.	LiDAR scans; photogrammetry models; terrain models
Time-Series / Sensor	Repeated measurements at fixed points over time, often streamed in real time.	Rainfall gauges; river-level telemetry; IoT and weather stations
Location / Trajectory	Sequential coordinates tracing how something moves over time.	GPS traces; mobility data; vehicle or vessel tracks

Data type	What it is	Examples
Tabular / Attribute	Structured records linked to a location, ready to join onto a map.	Census statistics; land-use records; socioeconomic indicators

Practical rule of thumb: raster answers "what does the surface look like?", vector answers "where exactly are the things?", time-series answers "how is it changing right now?", and tabular answers "who and what is there?". Your Activity 1 canvas should say which of these your idea needs.

3. Applications of GeoAI — Four Domains, Four Malaysian Case Studies

GeoAI is applied across four headline domains. Each is illustrated below with a real, documented Malaysian case study, so you can see the challenge, the technical approach, the measured impact, and the exact data types used.

3.1 Climate — Satellite AI for Deforestation-Free Palm Oil (Sabah & national sector)

The challenge. Malaysia's palm oil sector needed to prove its supply chains are deforestation-free to keep EU market access once the EU Deforestation Regulation (EUDR) took effect on 30 December 2025, while tracking carbon-relevant land-use change nationally.

The GeoAI approach. Companies pair high-resolution satellite imagery (e.g. the free European Sentinel-2 constellation, which revisits every location every ~5 days) with AI-powered change-detection monitoring to track land-use change on and around plantations over time, flagging suspected deforestation events for human verification and building an auditable, traceable evidence trail for regulators and buyers.

Impact. 5.5 million ha of Malaysian oil palm concessions are now trackable via satellite monitoring platforms; Malaysia has maintained a lower deforestation rate than regional peers since its 2016 peak; EU market access is maintained for compliant growers, along with improved sustainability credentials.

Data types used: optical satellite imagery (raster) · multispectral change data · plantation boundaries (vector). GeoAI tasks: change detection + pixel classification.

3.2 Smart Cities — Modelling Kuala Lumpur's Urban Heat Island

The challenge. Kuala Lumpur has lost over 68% of its vegetation cover in 32 years while built-up land expanded by more than 50%, intensifying the urban heat island (UHI) effect — cities being measurably hotter than surrounding rural areas because concrete and asphalt absorb and re-radiate heat — and raising heat-related health and energy risks for residents.

The GeoAI approach. Researchers combined satellite-derived Land Surface Temperature (LST) with ground weather-station data and Random Forest machine learning. Crucially, they used SHAP (SHapley Additive exPlanations — a technique that quantifies how much each input contributed to a prediction) to explain which factors drive heat exposure, making the model interpretable rather than a black box.

Impact. A 1.5–1.7°C UHI intensity increase has been recorded in KL/Petaling Jaya over recent decades; DBKL has planted 200,000+ trees under a data-informed green-infrastructure and tree-tracking programme, with 5,000+ AI-enabled smart cameras feeding its city command centre.

Data types used: satellite land-surface temperature (raster) · weather-station time-series · building/land-use (vector). GeoAI tasks: prediction/regression + pattern mining.

3.3 Agriculture — AI Crop Health Monitoring on a Sabah Plantation

The challenge. A Sabah-based palm oil producer needed to catch pest outbreaks, disease, and irrigation or fertiliser problems early across vast plantation areas that are impractical to inspect on foot at scale.

The GeoAI approach. Satellite imagery is paired with AI-powered monitoring that computes vegetation indices such as NDVI (Normalised Difference Vegetation Index — a standard measure of plant health calculated from how strongly vegetation reflects near-infrared vs red light; healthy plants score high, stressed plants score low). The system automatically flags vegetation stress, counts individual palm trees, and generates alerts that trigger targeted field inspections instead of routine blanket checks — a practice known as precision agriculture.

Impact. Earlier detection of pest attacks, disease, and nutrient or water stress across the estate; more targeted use of fertiliser, pesticide, and water — cutting both cost and environmental impact; yield data feeds drought monitoring and climate-adaptation planning for future seasons.

Data types used: satellite and drone imagery (raster) · vegetation index (NDVI) · plantation parcels (vector). GeoAI tasks: object detection (tree counting) + change detection.

3.4 Disaster Management — Deep Learning Flood Forecasting for the Klang River

The challenge. Malaysia's intensifying monsoon rainfall raises flood risk in dense river basins like the Klang River, where fast-rising water leaves communities little time to prepare or evacuate.

The GeoAI approach. Researchers integrated deep learning models — specifically LSTM (Long Short-Term Memory) neural networks, a type designed for sequences over time, which makes them ideal for rainfall and river-level series — with GIS-based catchment and rainfall data to forecast river flow and flood risk. These feed Malaysia's national telemetry- and radar-based flood forecasting and warning system run by JPS (Jabatan Pengairan dan Saliran — the Department of Irrigation and Drainage).

Impact. Hours of additional lead time for at-risk communities versus manual forecasting; global benchmarks suggest effective early-warning systems can cut flood-related fatalities by up to 43% and flood-related economic costs by 35–50%.

Data types used: rainfall and river-level time-series · radar precipitation (raster) · river catchment (vector/GIS). GeoAI tasks: time-series prediction.

Notice the pattern across all four cases: (1) a real problem too big or too fast for manual work; (2) a combination of two or more data types; (3) an AI model suited to that data; (4) predictions turned into a decision someone can act on. Your hackathon project should be able to tell the same four-part story.

4. The Geospatial-AI Innovation Canvas

The GeoAI Canvas is the stepped methodology you will use to structure the innovative and collaborative narrative of your hackathon project. It is grounded in cognitive learning theories (Cognitive Structural Theory, Cognitive Complexity Theory, Constructivist Learning) and innovation theory (Computational Thinking, Systems Design Thinking, Product Design). Its key fields:

- **Pains —** the real frustrations and problems your target users face.
- **Gains —** what they want, need or dream of; the benefits they seek.
- **Jobs to be Done —** what is on their "to solve / to do" list.
- **Promise —** how the solution creates value: does it alleviate pains and create gains?
- **Experience Journey —** the chronological touchpoints before, during and after, and how they shape behaviour change (Entering Behaviour → Exiting Behaviour).
- **Sustainable Commitment & Return —** what stakeholders commit to sustain the project, and what they walk away with (e.g. certification as a Certified Geospatial-AI Innovator, CGAI²).
- **Cost, Revenue & Satisfaction —** fixed/variable costs, potential revenue streams, and how participants would rate and describe the experience.

Keep the Canvas visible from Day 1 — its Stakeholder field feeds directly into the stakeholder-engagement techniques in Session 2 and the ethics work in Sessions 3–6.

5. Why Innovation and Collaboration Matter

No single discipline can solve complex geospatial problems alone. GeoAI innovation happens at the intersection of domain experts, data scientists, ethicists, and the communities affected.

- **Cross-disciplinary teams —** geographers, engineers, and social scientists catch blind spots any one field would miss.
- **Rapid iteration —** hackathon formats reward fast prototyping, testing, and honest feedback loops.
- **Community-centred design —** solutions endure when the people affected help shape them from the start.

The Innovation Mindset

Innovation is deliberately generating more, and better, options than the first idea that comes to mind — then testing them against reality.

- **Diverge, then converge:** generate many possible solutions before narrowing to one — resist the pull of the first idea.
- **Fail fast, learn faster:** a hackathon prototype is a question, not a final answer — cheap failure buys fast learning.
- **Cross-pollinate ideas:** borrow methods from other fields; a healthcare or logistics technique may solve a mapping problem.
- **Sit with ambiguity:** real problems rarely arrive well-defined — innovators get comfortable starting before everything is known.

The Collaboration Model

Collaboration is structured teamwork across differences — of discipline, seniority, and lived experience — so blind spots get caught before launch, not after.

- **T-shaped teammates:** deep in one skill, conversant in many — able to work fluently across a data scientist, a designer, and a domain expert.
- **A shared vocabulary:** agree early on what “accuracy”, “risk”, or “done” mean — most team friction is a definitions problem in disguise.
- **Open data & documentation:** work others can pick up, audit, and extend outlives the hackathon weekend.
- **Communities of Practice (CoP):** the Canvas leans on CoP networks — collaboration that continues after the event ends, including regional networks across ASEAN member states.

Remember: innovation without collaboration stays a prototype on someone’s laptop; collaboration without innovation just maintains the status quo, together. Hackathon teams need both.

6. Hands-On Activity 1 — GeoAI Around Us (10 minutes)

Task. Individually or in pairs, identify a real-world problem in your community and propose a possible AI-powered solution. If you are studying these notes alone, do this exercise on paper — it is the foundation for everything in Sessions 2–6.

6. Spot a real problem you observe locally (e.g. flooding, traffic, land use in your own state or hometown).
7. Identify what geospatial data could describe it (use the six data types in Section 2).
8. Propose an AI/ML approach to address it (use the common tasks table in Section 1).

Worked example: Flood prediction → rainfall data + satellite images → machine learning model.

Deliverable: a completed Problem–Data–Solution Canvas.

The Problem	The Data	The AI Solution
Recurring flash floods in low-lying neighbourhoods (example)	Rainfall records + satellite imagery + river sensor data (example)	ML model forecasting flood risk 6–12 hrs ahead (example)
Your answer:	Your answer:	Your answer:

Glossary

Term	Meaning
GIS	Geographic Information System — software for storing, managing, analysing and mapping spatial data.
Remote sensing	Measuring the Earth's surface from a distance, usually via satellites or drones.
Sentinel-2	A free European Space Agency satellite constellation providing multispectral imagery of the whole Earth roughly every 5 days.
NDVI	Normalised Difference Vegetation Index — a plant-health score computed from near-infrared vs red reflectance.
LiDAR	Light Detection and Ranging — laser scanning that produces precise 3D point clouds of terrain and structures.
LSTM	Long Short-Term Memory — a neural network type designed for sequences over time (e.g. rainfall series).
Random Forest	A classic machine-learning method that combines many decision trees for robust predictions.
SHAP	SHapley Additive exPlanations — a technique that quantifies how much each input contributed to a model's prediction.
EUDR	EU Deforestation Regulation — requires proof that commodities sold into the EU are deforestation-free (in effect 30 Dec 2025).
UHI	Urban Heat Island — the phenomenon of cities being measurably hotter than surrounding rural areas.
Pretrained model	A ready-made deep-learning model already trained for a task (e.g. building extraction), usable without collecting training data.
Foundation model	A very large model pretrained on massive diverse data, adaptable to many downstream tasks by fine-tuning.

Self-Check Questions

1. In one sentence each: what does traditional GIS do, and what does GeoAI add on top of it?
2. Your team wants to count every rooftop water tank in a city from aerial photos. Which GeoAI task is this, and which data type(s) would you need?
3. Why were LSTM networks (rather than a simple average of past readings) chosen for Klang River flood forecasting?
4. Name the four elements every strong GeoAI case study has in common (hint: see the note at the end of Section 3).
5. Which two fields of the GeoAI Canvas most directly feed the ethics work in later sessions, and why?

Key Takeaways

- GeoAI blends AI with location data to detect patterns, predict risk, and guide real decisions at scale — going beyond GIS's manual interpretation.
- Under the hood, GeoAI runs on machine learning and deep learning, following a Collect → Prepare → Train → Predict → Decide loop; pretrained models mean you rarely start from zero.
- The six data types (raster, vector, point cloud, time-series, trajectory, tabular) define what your model can and cannot do.
- Malaysia already runs serious GeoAI at national scale — palm oil EUDR compliance, KL heat mapping, plantation monitoring, and Klang River flood forecasting.
- The GeoAI Canvas gives your project a structured, stakeholder-centred narrative from Day 1.
- Innovation and collaboration are inseparable: diverse teams plus rapid iteration produce solutions that survive contact with reality.

Sources & Further Reading

- Esri — What is GeoAI? (esri.com/en-us/capabilities/geoai/overview) and Demystifying GeoAI (ArcNews).
- Esri ArcNews (2026) — GeoAI in the Age of Foundation Models; A Quick-Start Guide to Esri's Pretrained GeoAI Models.
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- Machine Learning in Modeling Urban Heat Islands — Kuala Lumpur (Springer, 2025); OpenGov Asia — DBKL Smart City Vision (2025).
- Deep Learning-Based Forecast and Warning of Floods in Klang River, Malaysia (IIETA); JPS — National Flood Forecasting & Warning System.